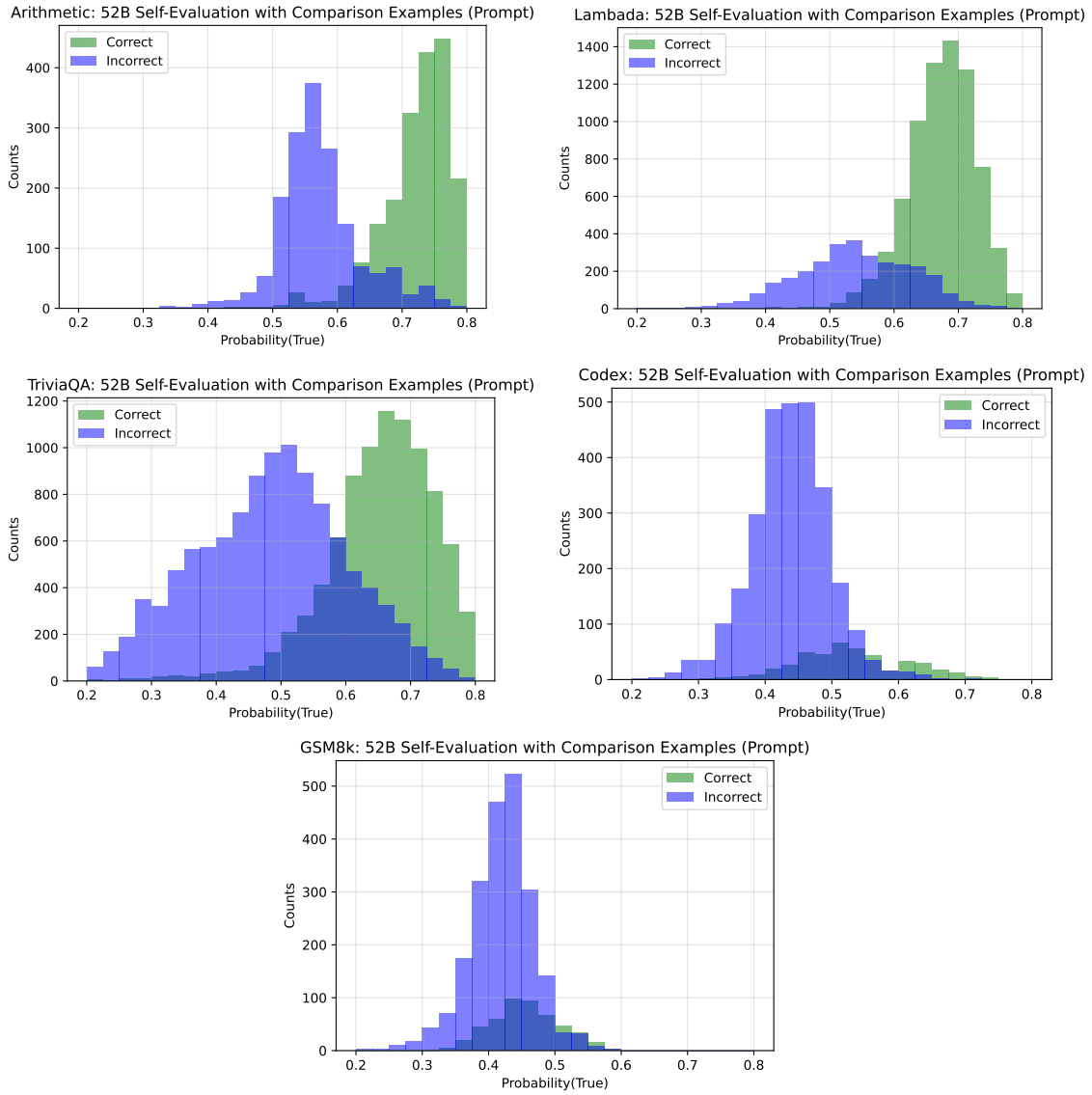

Figure 38 Models self-evaluate their samples by producing a probability P(True) that the samples are in fact correct. Here we show histograms of P(True) for the correct and incorrect samples for each evaluation, in the evaluation paradigm where models also get to see a total of five $T = 1$ samples for the same question, in order to improve their judgment. P(True) provides very noticeable separation between correct and incorrect samples with the 52B model, but the probability itself is not calibrated. In particular, almost all samples from arithmetic questions have P(True) > 0.5. These evaluations are all zero-shot, but they include a human-written prompt with six examples.

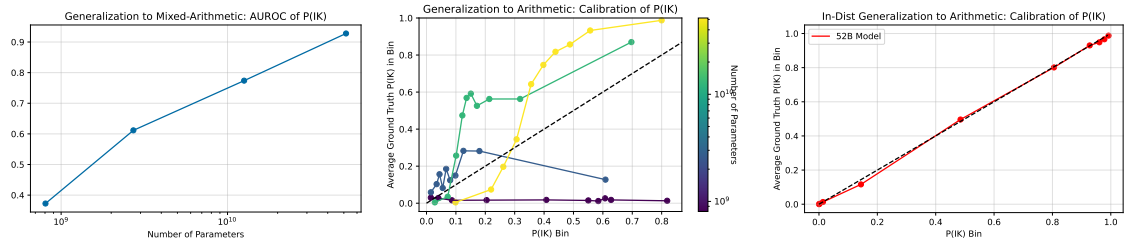

Figure 39 Left and Middle: Generalization of P(IK), trained on TriviaQA, to Arithmetic. Right: Calibration curve of a 52B model that was trained for P(IK) on all 4 tasks, on Arithmetic

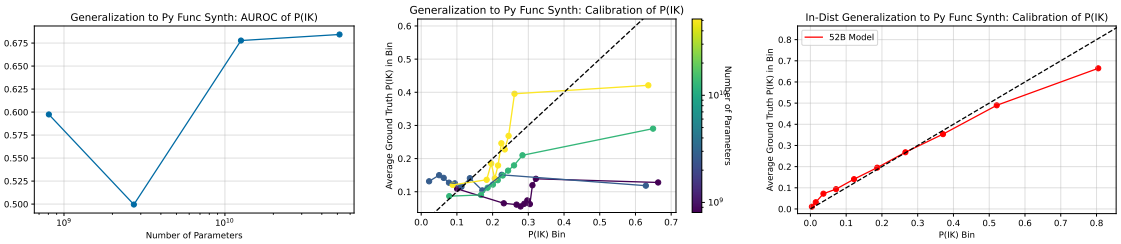

Figure 40 Left and Middle: Generalization of P(IK), trained on TriviaQA, to Python Function Synthesis. Right: Calibration curve of a 52B model trained for P(IK) on all 4 tasks, on Python Function Synthesis.

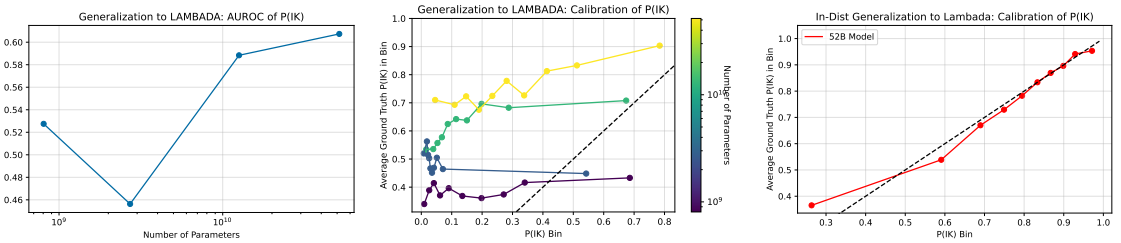

Figure 41 Left and Middle: Generalization of P(IK), trained on TriviaQA, to LAMBADA. Right: Calibration curve of a 52B model that was trained on all 4 tasks, on LAMBADA. Note calibration is extremely poor because P(IK) is systematically too small for all predictions; see Figure 16.

Name of Mixed-Arithmetic Subset	Example Question and Answer
add_digits_1_nlp_comma	Question: What is 4 plus 9?\n\nAnswer: 13
add_digits_2_nlp_comma	Question: What is 13 plus 88?\n\nAnswer: 101
add_digits_3_nlp_comma	Question: What is 423 plus 740?\n\nAnswer: 1,163
add_digits_4_nlp_comma	Question: What is 5,476 plus 4,136?\n\nAnswer: 9,612
add_digits_5_nlp_comma	Question: What is 95,837 plus 27,060?\n\nAnswer: 122,897
multiop_digits_1_nlp_comma	Question: What is 5 * 1 - 6?\n\nAnswer: -1
multiop_digits_2_nlp_comma	Question: What is 65 * 81 + 44?\n\nAnswer: 5,309
multiop_digits_3_nlp_comma	Question: What is 622 * 363 - 596?\n\nAnswer: 225,190
multiply_digits_1_nlp_comma	Question: What is 4 times 9?\n\nAnswer: 36
multiply_digits_2_nlp_comma	Question: What is 13 times 88?\n\nAnswer: 1,144
multiply_digits_3_nlp_comma	Question: What is 423 times 740?\n\nAnswer: 313,020
multisum_digits_1_nlp_comma	Question: What is 8 + 2 + 5 + 7 + 6?\n\nAnswer: 28
multisum_digits_2_nlp_comma	Question: What is 38 + 82 + 43 + 19 + 45?\n\nAnswer: 227
multisum_digits_3_nlp_comma	Question: What is 975 + 871 + 145 + 704 + 397?\n\nAnswer: 3,092
subtract_digits_1_nlp_comma	Question: What is 4 minus 9?\n\nAnswer: -5
subtract_digits_2_nlp_comma	Question: What is 13 minus 88?\n\nAnswer: -75
subtract_digits_3_nlp_comma	Question: What is 423 minus 740?\n\nAnswer: -317
subtract_digits_4_nlp_comma	Question: What is 5,476 minus 4,136?\n\nAnswer: 1,340
subtract_digits_5_nlp_comma	Question: What is 95,837 minus 27,060?\n\nAnswer: 68,777

Table 4 Example questions and answers from our programmatically generated Mixed-Arithmetic dataset.

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